LINWEI CHEN 陈林蔚



TEL: +86 18910814867 · +852 46344786
Homepage: https://linwei-chen.github.io/
WeChat: Uno-Whoiam· Age: 30 · chenlinwei.ai@gmail.com

Research Interests: Frequency-domain Representation Learning, Dense Prediction, Degraded & Scientific Imaging, Medical AI (Foundation Models & LLMs)

EDUCATION

The University of Hong Kong, School of Computing and Data Science, *Post-Doctoral Fellow*
(Supervisor: Prof. Yizhou Yu) 2026.9 - Present

The Hong Kong University of Science and Technology, HKGAI, *Post-Doctoral Fellow*
(Supervisor: Prof. Yizhou Yu) 2026.1 - 2026.8

Beijing Institute of Technology, School of Computer Science, Electronic Information, *Ph.D.*
(Supervisor: Prof. Ying Fu) 2021.9 - 2025.6

Beijing Institute of Technology, School of Computer Science, Software Engineering, *M.S.*
(Supervisor: Prof. Ying Fu) 2019.9 - 2021.6

China University of Geosciences (Beijing), School of Engineering and Technology, Mechanical Design, Manufacturing and Automation, *B.S.* 2014.9 - 2019.6

SELECTED PUBLICATIONS

1. **Linwei Chen**, Lijun Liu, Ying Fu*, Meng Tian, Zhishou Zhang, Hengfu Cui, Qiang Ju, Qianxi Li, Jungong Han*, Yizhou Yu*, and Hong-Yu Zhou*. An open AI system for dermatology tasks. **Nature Biomedical Engineering** (**IF=26.3**, Nature family), **Under Major Revision**. (First author. A fully open dermatology foundation model trained on public data, matching private-cohort systems; from representation learning to clinical translation.)
2. Shuai Wu†, **Linwei Chen**† (Co-first), …, Tien Yin Wong, Yizhou Yu, and Hong-Yu Zhou. The accuracy-safety trade-off of large reasoning models in clinical diagnostics. **Nature Communications** (**IF=18.1**), **Accepted**.
3. **Linwei Chen**, Ying Fu*, Lin Gu, Chenggang Yan, Tatsuya Harada, and Gao Huang. Frequency-aware Feature Fusion for Dense Image Prediction. IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**). (A generalized optimization approach for feature fusion representation, easily applicable to medical images. **IF=20.4**, **CCF-A**, Web of Science Citation Top 0.3%) [Paper] [Code]
4. **Linwei Chen**, Ying Fu*, Lin Gu, Dezhi Zheng, Jifeng Dai. Spatial Frequency Modulation for Semantic Segmentation. IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**). (Optimization of downsampling. **IF=20.4**, **CCF-A**) [Paper] [Code]
5. **Linwei Chen**, Ying Fu*, Kaixuan Wei, Dezhi Zheng, Felix Heide. Instance Segmentation in the Dark. International Journal of Computer Vision (**IJCV**) 2023. (Tackling extremely high noise; understanding complex scenes under low-light conditions. **IF=10.3**, **CCF-A**) [Paper] [Code & Dataset]
6. **Linwei Chen**, Lin Gu, Dezhi Zheng, Ying Fu*. Frequency Adaptive Dilated Convolution for Semantic Segmentation. IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2024. (**Highlight**, top 2.8%, Optimized dilated-convolution representation, applicable to medical images. **CCF-A**) [Paper] [Code]
7. **Linwei Chen**, Lin Gu, Liang Li, Chenggang Yan, Ying Fu*. Frequency Dynamic Convolution for Dense Image Prediction. IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2025. (Optimization of dynamic convolution, easily applicable to VLM. **CCF-A**) [Paper] [Code]
8. **Linwei Chen**, Lin Gu, Ying Fu*. Frequency-Dynamic Attention Modulation for Dense Prediction. International Conference on Computer Vision (**ICCV**) 2025. (Optimization of attention mechanism representation in Transformer architecture, easily applicable to VLM. **CCF-A**) [Paper] [Code]
9. **Linwei Chen**, Lin Gu, Ying Fu*. When Semantic Segmentation Meets Frequency Aliasing. The International Conference on Learning Representations (**ICLR**) 2024. (Analysis of hard pixel samples. **Class A** top conference in artificial intelligence) [Paper] [Code]

HONORS, AWARDS & ACADEMIC SERVICES

1. 2024 China National Scholarship for Doctoral Students, Ministry of Education, China; 2022 BIT Doctoral Special Scholarship
2. CVPR 2024 PBDL International Challenge, Low-Light Track: 2nd Place (Segmentation), 3rd Place (Detection); 2021 “Fire Eye” AI Challenge, Top 8 / 500+ Teams Nationally
3. Reviewer: TPAMI / TIP / IJCV / MedIA; CVPR / ICCV / ECCV / NeurIPS / AAAI / BMVC. Outstanding Reviewer of CVPR 2026 & BMVC 2024. Guest Editor: Journal of Imaging. Patents: 8 filed (3 granted).